MIL-C-45956(WC)

23 July 1974
SUPERSEDING
(See Section 6)

MILITARY SPECIFICATION

CONTROLLER, WEAPONS-GUN SPEED AND LAUNCHER BRAKE, LH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of weapons controller for installation in the left hand position of the M28A1 Armament Subsystem. The left weapons controller contains the required circuity to control firing speed of the left machine gun, M134 or the drive and brake of the left M129 Grenade Launcher.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-W-13855	-	Weapons, Small Arms and Aircraft Armament Subsystems, General Specification for.
MIL-P-14232	-	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of.
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of.
MIL-A-45967	-	Armament Subsystem, Helicopter: M28A1

FSC 1090

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STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.

DRAWINGS

Rock Island Arsenal

F11688531	-	Controller, Weapons-Gun Speed and Launcher Brake, (LH).
IEL11688531	-	Inspection Equipment List.

(Product drawings referenced in this specification form a part of F11688531; acceptance inspection and test equipment drawings form a part of IEL1168853.)

PUBLICATIONS

Rock Island Arsenal

P11688531	-	Packaging Data Sheet for Controller, Weapons-Gun Speed and Launcher Brake, (LH).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.2). Unless otherwise specified (see 6.2), the first article shall include a pilot pack (see 5.1).

3.2 Materials, construction and design. Controllers shall conform to the material, construction and design requirements specified herein, on Drawing F11688531, and in MIL-W-13855.

3.2.1 Weight. The weight of the controller shall not exceed 8.5 pounds.

3.3 Functional requirements.

3.3.1 Power input. The controller shall be capable of meeting the requirements specified herein, when supplied with 22 to 30 volts direct current (vdc) and a maximum rate of 3.5 amperes (amps).

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3.3.2 Weapon clearing (7.62mm). The weapon clearing function shall be performed utilizing the following relays:

A7K1	Time delay relay (7.62mm)
A7K3	Unclear relay
A7K6	Gate relay
A7K7	7.62mm Low speed relay
A7K8	Clear/Unclear memory relay
A7K9	7.62mm High/Low speed relay

Relays A7K10, A7K12, connector A7J1 and pin E of connector A7P1 shall be energized when 22-30 vdc is applied to pins B, R and E of connector A7P1 and to A7J1. Any references to a circuit being energized shall assume the grounded (negative) side of the power supply is connected to pin V of connector A7P1.

3.3.2.1 Unclear mode (7.62mm). Relay A7K3 shall be energized when 22-30 vdc is applied to pin N of connector A7P1, thereby actuating all A7K3A and A7K3B contacts. Applying 22-30 vdc to pin C of connector A7P1, relays A7K1, A7K7, A7K8 (clear memory coil) and A7K9 (low speed coil X1-X2) shall be energized through the normally closed contacts A7K11A and closed A7K12B. Once relay A7K1 is energized, relays A7K6 and A7K8 (unclear memory coil) shall become energized through A7K1A contacts and power from pin E of connector A7P1. Relay A7K8 will stay latched in the clear memory position, contact A7K8B will be open and no voltage shall exist at pin P of connector A7P1. At this time, voltage shall exist at pin C of connector A7P2 and continuity shall exist between pins D and E of connector A7P2. Upon removal of power at pin C of connector A7P1, relays A7K7 and A7K8 (clear memory coil), and A7K9 (low speed coil X1-X2) shall immediately deenergize. Relay A7K7 and A7K9 (low speed) contacts shall open removing power from pin C of connector A7P2 and cause an open circuit between pins D and E of connector A7P2. Relay A7K6 and the unclear memory coil of relay A7K8 shall remain energized from power applied to pin E of connector A7J1 and by the time delay action of relay A7K1 for a-period of 250 + 40 milliseconds. Upon completion of the time delay period, relay A7K6 and A7K8 (unclear memory coil) shall be deenergized. Because of the latching action, relay A7K8 shall remain in the unclear memory position. Relay contact A7K8B shall be closed at the time power is removed from pin C of connector A7P1. (Power applied to pin P of connector A7P1 from 22-30 vdc on pin R of connector A7P1 and diode CR12). A7K6A contacts shall remain closed during the time relay A7K1 is closed, and during the delay period. Continuity shall exist between pins F and G of connector A7P1.

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3.3.2.2 Clear mode (7.62mm). Relay A7K3 shall not be energized (A7K3 contacts set in normal position) when 22-30 vdc is applied to pin C of connector A7P1, thereby energizing relays A7K1, A7K6 and A7K8 (unclear memory coil). When relay A7K1 becomes energized, relays A7K7, A7K8 (clear memory coil), and A7K9 (low speed coil X1-X2) shall become energized through relay contacts A7K1 and A7K3B with power on pin E of A7P1. Relay A7K8 will stay locked in the unclear memory position. Contact A7K6A shall close when power is applied to pin C of A7P1 resulting in continuity between pins F and G of A7P1. A7K8B shall close at the same time providing power to pin P of A7P1. Pin C of A7P2 shall be energized from A7J1 through A7K9A1, A7K9A2 and A7K7A contacts. Continuity shall exist between pins D and E of A7P2. When power is removed from pin C of A7P1, relay A7K6 and A7K8 (unclear memory coil) shall deenergize. This will latch relay A7K8 in the clear memory position where it shall remain energized for completion of the time delay of relay A7K1. Relay A7K1 (time delay) shall maintain contact with A7K1A (closed for a period of 250 + 40 milliseconds), thereby providing energy to relays A7K7 and A7K9 (low speed coil X1-X2), and the clear memory coil of relay A7K8 until completion of the time delay period of relay A7K1. Upon completion of the time delay period, relay A7K8 shall remain in the clear memory position due to latching action. Contacts A7K8B shall be open, and there shall be no voltage on pin P of A7P1 from pin R of A7P1. Contact A7K6A shall open upon removal of power from pin C of A7P1. There shall be an open circuit between pins G and F of A7P1. At the end of the delay period, contacts A7K9A1, and A7K9A2 and A7K7A2 shall open, removing power from pin C of A7P2. An open circuit shall exist between pins D and E of A7P2.

3.3.3 Low speed operation (7.62mm). This function shall be performed by the following relays:

A7K1	Time delay relay (7.62mm)
A7K6	Gate relay
A7K7	7.62mm Fire relay
A7K9	7.62mm High/Low speed relay

For the low speed operation, power shall be applied to the same pins as required in the weapon clearing functions (see 3.3.2). The same relay functions shall apply in either the clear or unclear memory condition. When 22-30 vdc is applied to pin C of A7P1, relays A7K7 and A7K9 (low speed coil) shall be energized. The contacts of relay A7K7A2, A7K9A1, A7K9A2, A7K9B1 and A7K9B2 shall close. The contacts of relay A7K7A2, A7K9A1 and A7K9A2 shall cause a redundant current path from A7J1 to pin C of A7P2. Contacts of relay A7K9B1 and A7K9B2 shall cause a current path from pin D to pin E of A7P2. Relay A7K6A shall close to provide continuity between pin G and pin F of A7P1.

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3.3.4 High speed operation (7.62mm). This function shall be performed by the following relays:

A7K1	Time delay relay (7.62mm clear)
A7K6	Gate relay
A7K7	7.62mm Fire relay
A7K9	7.62mm High/Low speed relay
A7K11	7.62mm High speed relay

For the high speed operation, power shall be applied to the same pins as required in the weapon clearing functions (see 3.3.2). The same relay functions shall apply in either the clear or unclear memory condition. When 22-30 vdc is applied to pin D of A7P1 all relay functions shall remain the same except that relay A7K11A contacts shall be open and relay A7K9 (low speed coil) shall not be energized. Relay A7K9 (high speed coil) shall be energized from power on pin D of A7P1 through A7K12A contacts. With relay A7K9 (high speed coil) energized, A7K9 contacts A3, A4, B3 and B4 shall be closed, allowing a current path between pin E and pin F and between pin C and pin D of A7P1. Relay A7K7A contacts shall remain closed and a current path shall exist between A7J1 and pin C of A7P1.

3.3.5 Weapon enable. The weapon enable relay shall be energized by applying 22-30 vdc to pin B of A7P1. Relay contact A7K10A shall close providing a circuit to relay A7K1 and associated relays, to provide power to either the high or low speed coils of relay A7K9. This shall close relay A7K9A or A7K9B contacts and provide a current path from A7J1 to pin C of A7P2, between pin D and pin E of A7P2 or between pin C and pin D and between pin E and pin F of A7P2. Relay contact A7K10B shall open to provide an open circuit between pin L and pin A of A7P1. Prior to energizing relay A7K10, a closed circuit existed from pin L to pin A of A7P1 through the A7K10B contacts and diode CR13. Relay contacts A7K10C shall close to provide power from pin M of A7P1 to relay A7K2 and A7K5 (relay coil). Relay contacts A7K10D shall be closed and continuity shall exist between pins W and S of A7P1.

3.3.6 40mm Gun drive. This function shall be performed by the following relays:

A7K2	Time delay relay (40mm clear)
A7K5	Drive latch relay
A7K9	40mm Drive brake relay

The 40mm gun drive function shall require 22-30 vdc applied to pins H, B and E of A7P1 and to A7J1. Power at pin B shall energize relay A7K10 which shall close contact A7K10C. 22-30 vdc applied to pin M of A7P1 shall energize relay A7K2 through contact A7K10C. Relay A7K5 shall become energized after contact A7K2A closes in the ground side of the A7K5 relay coil and power is applied on the plus side of the coil from

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pin M of A7P1, contact A7K10C and CR3. The A7K5A contacts shall close when relay coil A7K5 is energized forming a holding current for relay A7K5 when power is applied to pin H of A7P1. (Power on pin H is supplied from the 40mm magnetic brake sensor in the turret of the M28 subsystem). Contact A7K2B shall close upon energizing relay A7K2 and the power shall be supplied to pin G of A7P2. Contacts A7K5B1 and A7K5B2 shall close when relay A7K5 is energized, thus applying power to relay A7K9 (40mm drive coil) through the normally closed contact A7K12B. Relay A7K9 brake coil shall be deenergized since contacts A7K5B2, A7K5B3 are open. This will result in relay A7K9 being latched in the drive position with a current path from A7J1 to pin C of A7P2. Pin C of A7P2 shall be energized and continuity shall exist between pins D and E of A7P2 through contacts A7K9B1 and A7K9B2. With relay A7K9 latched in the drive position, power shall be applied to the A7K9 relay brake coil through A7K9A1, A7K9A2 contacts, CR8 diode, A7K12C and A7K12A. Relay A7K9 shall remain in the drive position since the drive coil was energized before the brake coil. Continuity shall exist between pin C of A7P2 and pin A of A7P3 and between pin F of A7P2 and pin D of A7P3.

3.3.7 40mm Gun braking. This function shall be performed by the following relays:

A7K2	Time delay relay (40mm clear)
A7K5	Drive latch relay
A7K9	40mm Drive/Brake relay

The 40mm gun braking function shall require 22-30 vdc applied to pins H, B, and E of A7P1 and to A7J1. Power to pin B shall energize relay A7K10 which will close contact A7K10C. When power is removed from pin M of A7P1, relay A7K2 will remain energized for 500 $\pm$ 50 milliseconds. All relays shall remain in the drive condition for 500 (ms) unless power is removed from pin H of A7P1. If power is removed from pin H, relay A7K5 will drop out of the drive condition immediately. At the end of the 500(ms) period or when power is removed from pin H of A7P1, relay A7K5 shall be deenergized. Removing power from the drive coil of relay A7K9 through the A7K5B1 and A7K5B2 contact openings will allow relay A7K9 to latch in the brake position for a period of 15(ms) or less. Relay A7K9 (brake coil) shall be energized from A7J1, contacts A7K9A1, A7K9A2, diode CR8, A7K12A and A7K12C normally closed contacts. Relay A7K9 brake coil shall be held in the brake position from power applied through relay contacts A7K5B2, A7K5B3 and pin E of A7P1. Power shall then be removed from pin G of A7P2 500 $\pm$ 50(ms) after relay A7K2 is deenergized. The action of relay A7K9 changing from the drive position to the brake position shall cause contacts A7K9A1, A7K9A2, A7K9B1 and A7K9B2 to open while contacts A7K9A3, A7K9A4, A7K9B3 and A7K9B4 close in 15(ms) or less. Diode CR9 connected from pin G to pin F shall dissipate the reverse voltage caused by the motor shunt field collapsing. Diode CR10 shall be connected from pins E to C of A7P2 to dissipate the energy from the back EMF generated while the motor armature is being dynamically braked.

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3.3.8 Diodes. Diodes reverse connected across A7K3, A7K6, A7K7, A7K10, A7K11 and A7K12 relay coils, shall dissipate the induced voltage generated in the collapsing relay coil field.

3.3.9 Condensers. The condenser connected across relay A7K1 shall absorb external switching voltage transients on pin C of A7P1, preventing relay A7K1 from being inadvertently energized.

3.4 Reliability. When specified in the contract (see 6.2), the controller shall have a minimum acceptable Mean-Mission-Between-Failure (MMBF) of not less than 1427 missions with a desired and specified MMBF of 2141 missions. A mission is defined as those functions of the controller necessary to simulate the expenditure of 4,000 rounds of 7.62mm ammunition and 265 rounds of 40mm ammunition during a two hour flight. A failure will be defined as a nonoccurrence of the appropriate event when power is applied and the required control switches are actuated. Testing shall be in accordance with 4.4.

3.4.1 Reliability criteria. The above reliability requirements are based on a minimum acceptable Mean-Time-Between-Failure (MTBF) of 2855 hours.

3.5 Environmental conditions. When specified in the contract (see 6.2), the weapons controller shall be capable of satisfying the environmental conditions in accordance with MIL-A-45967 for the system. When subjected to the environmental conditions, the weapons controller shall not exhibit any detrimental damage or deterioration which would prevent acceptable performance. Testing shall be in accordance with 4.4.

3.6 Electromagnetic interference (EMI). When specified in the contract (see 6.2), the weapons controller shall be capable of satisfying the EMI requirements in accordance with MIL-A-45967 for the system. When subjected to the EMI, the weapons controller shall not exhibit any detrimental damage or deterioration which would prevent acceptable performance. Testing shall be in accordance with 4.4.

3.7 Marking. Controllers shall be marked in accordance with the applicable drawing and MIL-W-13855. Serial numbers shall be assigned by the procuring activity (see 6.2).

3.8 Workmanship. Workmanship shall be in accordance with MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the

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performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. The first article (initial production unit(s) shall be submitted for inspection in accordance with contract requirements (see 6.2). The first article shall be representative of the production processes used during quantity production. The first article shall be subjected to quality conformance inspection specified herein, and when specified in the contract, reliability test (see 3.4 and 4.6.2), environmental test (see 3.5), electromagnetic interference (EMI) test (see 3.6, and such other inspections as is necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection.

4.5.1 Examination.

4.5.1.1 Component parts and concurrent repair parts. Examination of parts shall be performed in accordance with the criteria specified in the contract (see 6.2). Examination of parts shall be accomplished prior to their assembly into the end items or submission for acceptance as repair parts.

4.5.1.2 Inspection methods. The following provisions shall be applicable to the prescribed inspection methods. Request for a method other than that specified shall be submitted to the Government for approval.

a. Where "Visual" is specified as the inspection method for functioning assembly requirements, the assembly shall be visually examined for completeness and manually operated for functioning requirement.

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b. Where "SMTE" (Standard Measuring and Test Equipment) is specified as the method of inspection, the contractor may use any type of industry developed, commercially available, multi-usage equipment or special inspection and/or testing equipment approved by the Government.
c. Examination for workmanship shall include examination for missing operations and the presence of nicks, cracks, burs, scratches, toolmarks, deformations or other signs of poor workmanship. Controllers shall be carefully compared with a specimen of known acceptable quality and conform to the requirements of 3.8.

4.5.1.3 Controllers. Each controller shall be examined as specified below. The examination provisions should be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristics by subsequent operations. Reinspection of these characteristics is not required provided assurance exist that the characteristic has not been changed, degraded or damaged by subsequent manufacturing, assembly or handling and that adequate inspection records are maintained. Controllers failing to meet these requirements shall be rejected.

	Characteristic	Requirement	Inspection Method
a.	Missing, damaged, or incorrectly assembled component parts.	3.2	Visual
b.	Diodes and condensers connected across relay coils for proper installation and for electrical conductance while in the forward position and nonconductance while in the reverse position.	3.3.8 and 3.3.9	SMTE
c.	Illegible, incorrect or missing product markings and identification plates, serialization.	3.7	Visual
d.	Workmanship.	3.8	Visual

4.5.1.3.1 Weight. The first 3 controllers produced shall be examined for weight requirements (see 3.2.1).

4.5.2 Testing

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4.5.2.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete controller. If test requirements cited herein are not met, acceptance of the controller shall be deferred and the contractor shall accomplish as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional physical and visual examination of the components which are suspected to be the cause of failure or malfunction.
b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.
c. Examine controllers, partially assembled controllers and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned controller for retest.

4.5.2.2 Functional testing. Each controller shall be function tested (see 3.3) using the test method specified in 4.6.1. Controllers failing to meet the functional requirements shall be rejected.

4.5.2.3 Component parts and concurrent repair parts testing. Raw material testing, part testing, and certification shall be performed in accordance with the criteria specified in the contract (see 6.2). This will include chemical analysis and physical tests of materials, tests of protective finish, heat treatment, bonding, and function of parts as applicable. The contractor shall accomplish these tests prior to assembly of parts into the end item.

4.5.3 Packaging examination and testing. Unless otherwise specified (see 6.2), the packaging examination and testing shall be in accordance with MIL-P-14232.

4.5.4 Inspection equipment.

4.5.4.1 Acquisition, maintenance and disposition. Unless otherwise specified, (see 6.2) responsibility for acquisition, maintenance and disposition of acceptance and test equipment prescribed on the List of Inspection Equipment IEL11688531 and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

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4.5.4.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it must be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.6 Test methods.

4.6.1 Function test. Controllers shall be tested in accordance with the test procedure specified on IEL11688531 (ref. Drawing 11836856).

4.6.2 Reliability test. The reliability test shall be performed by repeating the test cycle defined in 4.6.2.1 to determine compliance with 3.4. For each test cycle, the failure versus test cycles data shall be cumulatively plotted. Testing shall be terminated whenever a decision point as defined in Figure 1 has been reached.

4.6.2.1 Reliability test cycle. One test cycle (representing the functions necessary to perform two missions) shall be completed by performing the following test methods contained in the functional test procedure, specified in IEL11688531 (ref. Functional Test Procedure 11836856), for the number of repetitions stated.

Functional Test Procedure Paragraph Number	Number of Repetitions
5.1	Initial only
5.2	Initial only
5.3	7
5.4	9
5.5	5
5.6	2
5.7	5
5.8	5
5.9	1
5.10	1
5.11	1
5.12	5
5.13	1
5.14	1
5.15	2
5.16	2
5.17	1
5.18	1

4.6.2.2 The contractor shall provide replacement parts as required to complete the test.

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4.6.2.3 At the option of the contractor, the test may be interrupted during, or at the end of, a cycle and the test restarted later at the point at which it was interrupted.

4.6.2.4 Complete, accurate records shall be kept for each reliability test, showing each failure and part replacement, including an indication of when each failure occurred. In addition, complete, accurate records shall be kept for all time losses attributable to weapons controller, including the time for each stoppage.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. The pilot pack shall consist of a complete controller packaged in accordance with Packaging Data Sheet P11688531 for the level of protection specified in the contract (see 6.2), packed level C and forwarded in accordance with 3.1.

5.2 Preservation, packaging, packing and marking. Controllers shall be preserved, packaged, packed and marked in accordance with Packaging Data Sheet P11688531 for the level of protection specified in the contract (see 6.2).

5.3 Repair parts. Repair parts shall be preserved, packaged, packed and marked in accordance with the Packaging Data Sheet or other requirements and for the level of protection specified in the contract (see 6.2).

6. NOTES

6.1 Intended use. The controller is intended for use on, but not limited to the M28A1 Helicopter Armament Subsystem.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. List of drawings and specifications pertinent to the controller showing applicable revision dates.
c. Requirements for and number of controllers to be submitted for first article by the contractor (see 3.1 and 4.4 and 5.1).
d. Reliability, environmental and electromagnetic interference (EMI) requirements, when required (see 3.4, 3.5, 3.6 and 4.4).
e. Block of serial numbers required (see 3.7).
f. Criteria for examination and testing of component parts and concurrent repair parts (see 4.5.1.1 and 4.5.2.3).

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g. Packaging examination and testing, if different (see 4.5.3).
h. List of inspection equipment, responsibilities for acquisition, maintenance, and disposition of, if other than as specified (see 4.5.4.1).
i. Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).
j. Packaging instructions for repair parts (see 5.3).

6.2.2 Management control systems requirement. When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208 on the Management Control Systems Summary List, DD Form 1660.

6.2.3 Unless otherwise specified (see 6.2), the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Systems Summary List, DD Form 1660.

6.3 Relay functions. The following describes the general functions of relays specified in Section 3.

Symbol	Type	Function
A7K1	Time Delay	Machine gun clearing - The time delay relay A7K1 (7.62mm clear) shall provide for continued operation of the machine gun and accessories for a period of 250 milliseconds upon release of the trigger. The delay shall permit expenditure of rounds already in the weapon at the time of trigger release. Trigger release also results in loss of energy in the gate solenoid thereby interrupting ammo feed to the weapon during the delay period.
A7K2	Time Delay	Grenade launcher clearance - The time delay relay A7K2 (40mm clear) shall provide continued operation of the grenade launcher and accessories upon release of the trigger for a period of time sufficient to clear the weapon chamber of ammunition.

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Symbol	Type	Function
A7K3	Switching	Unclear relay - Operation controlled by a switch on the gunner control panel. The contact action provides an option to the gunner to automatically clear or not to clear the rounds from the machine gun and feed mechanism upon release of the trigger (see A7K8). The contact action with relay energized, shall provide for actuation of the gate solenoid on the machine gun and the unclear weapon indicator light on the control panel. The contact action, with the relay not energized, shall provide for clearing of the weapon by means of time delay relay A7K1 and firing relays A7K7 and A7K9 as well as by closure of the entrance to the machine gun chamber by loss of energy in the gate solenoid coil.
A7K5	Switching	Energized by 40mm trigger, in turn energizing "drive" coil of A7K9. Upon release of trigger A7K5 is held energized by grenade launcher magnetic recoil sensor until next recoil occurs. A7K5 then deenergizes and thereby energizes "brake" coil of A7K9, dynamically braking weapon.
A7K6	Switching	Gate relay - This relay shall introduce the 22-30 vdc battery busbar power to the coil of the gate solenoid thereby clearing the entrance to the machine gun chamber.
A7K7	Switching	7.62mm Firing relay - This relay shall be actuated by the 7.62mm trigger and shall provide the primary firing contact for the machine gun at high speed with a 100 ampere steady state inductive current rating.

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Symbol	Type	Function
A7K8	Switching	Clear/Unclear memory relay. Following release of the trigger, the contacts of this relay shall magnetically remain in the position determined by A7K3. The contacts of this relay shall provide current to the weapon unclear indicator light on the gunner control panel when the weapon was not cleared and will interrupt this current the next time the weapon is cleared.
A7K9	Switching	40mm Brake/Drive relay or 7.62mm High/Low speed relay - This relay shall be energized either by action of relay A7K5 or by 7.62mm trigger action. With a 40mm weapon installed, A7K9 provides contact action which will introduce the 22-30 vdc battery busbar power directly into the grenade launcher drive motor. The contact action of this relay shall result in either starting or braking of the grenade launcher assembly and accessories. With a 7.62mm weapon installed, the 7.62mm trigger and A7K11 provide the switching necessary for A7K9 coils, either directly or through A7K1 contacts to achieve low speed, and directly as well as by action of A7K11 contacts to achieve high speed. The contacts of this relay switch the 7.62mm drive motor armatures to either a parallel or series configuration, providing high and low speeds. These contacts should have a 50 ampere steady state inducive current rating. Relay A7K12 determines whether A7K9 acts as a 40mm or 7.62mm firing relay.
A7K10	Switching	Weapon enable relay - This relay shall be energized by the weapon select switches on the gunner sight and pilot control panel. The contact action shall place the controller in a standby condition which will permit armament operation upon actuation of the trigger. This relay prevents firing of an unselected weapon.

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Symbol	Type	Function
A7K11	Switching	7.62mm High speed relay - The second detent of the 7.62mm trigger shall actuate this relay, while at the same time energizing the high speed coil of A7K9. The contacts of this relay shall interrupt the current to the low speed coil of A7K9 allowing the gun to fire at high rate of speed.
A7K12	Switching	7.62mm Weapon installed relay - When a 7.62mm weapon is installed, this relay shall be energized off the turret power bus through a jumper wire in the 7.62mm chorting connector (A3A1P2 or A3A2P2). The contacts of this relay shall be used as an interlock between the 7.62mm and the 40mm firing circuits by setting up the coil circuits of A7K9 to receive 7.62mm or 40mm trigger signals.

6.4 Firing interlock. With the 7.62mm weapon installed, relay A7K12 shall be energized by 22-30 vdc applied to pin R of A7P1. The contacts of this relay provide for an interlock between the 7.62mm and 40mm firing circuits to prevent the firing of the unselected weapon.

6.5 Supersession data. This specification includes the requirements of U.S. Army Weapons Command Purchase Description, USAWECOM PD-30, Amendment 1, dated 31 March 1972.

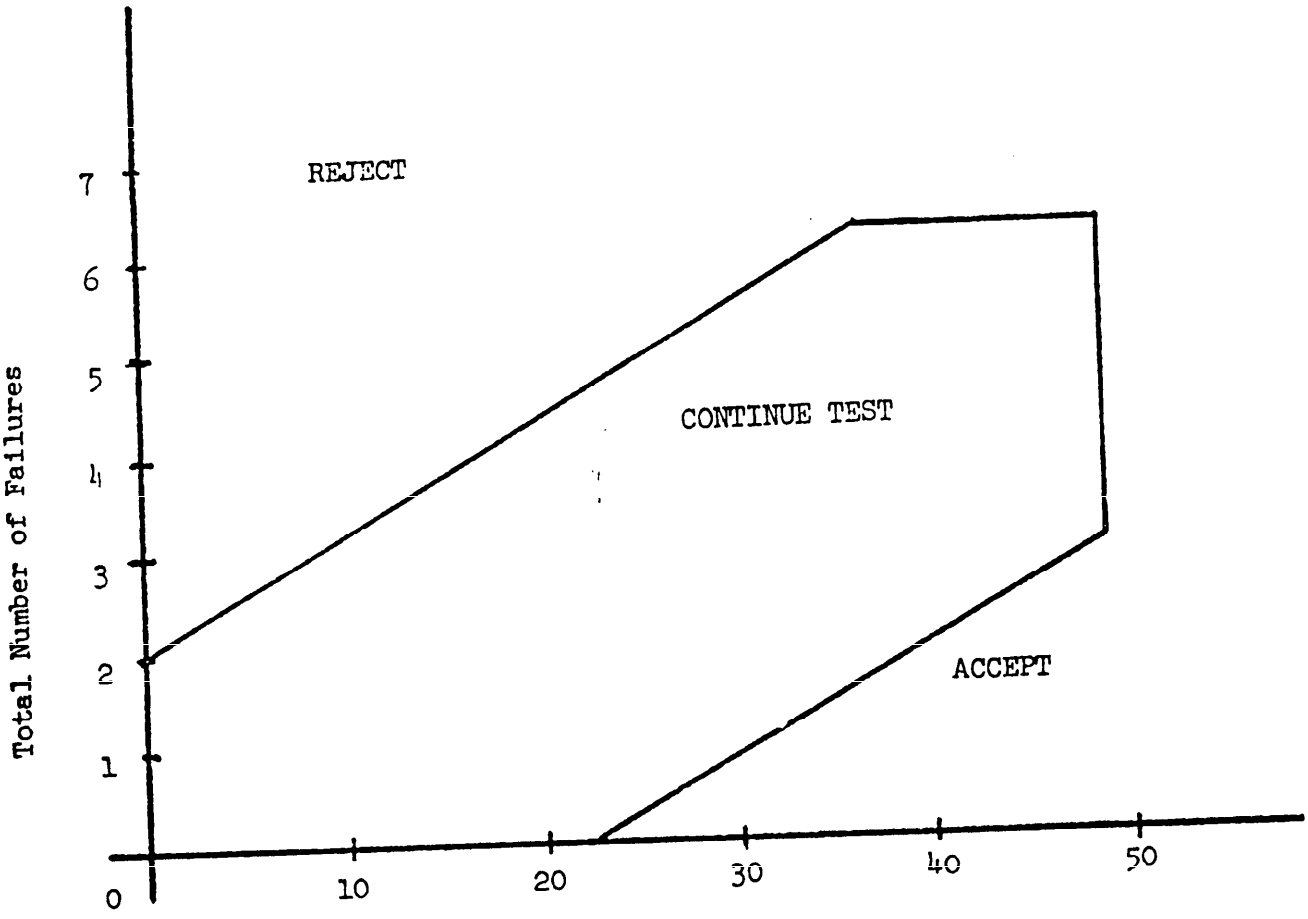
Custodians:
Army - WC

Preparing activity:
Army - WC

Project number:
1010-A067

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Decision Risks: 30%
Discrimination Ratio: 1.5:1
MIL-STD-781B Test Plan VII



NO. OF FAILURES	REJECT (EQUAL OR LESS)	ACCEPT (EQUAL OR MORE)
0	N/A	2248
1	N/A	3115
2	N/A	3983
3	867	4850
4	1734	4850
5	2601	4850
6	4850	N/A

FIGURE 1

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